

Introduction to Automata Theory

Problem Set 2

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Some definitions

For the purposes of this problem set, assume the following definitions unless stated otherwise:

- **Minimal DFA/NFA:** An automaton recognizing a regular language L is considered minimal if it has the fewest possible states among all such automata recognizing L . (Note: While minimal DFAs are unique up to isomorphism, minimal NFAs are not necessarily unique).
- **ϵ -closure:** Let $S \subseteq Q$ be a set of states in an NFA. The ϵ -closure of S , denoted $E(S)$, is the set of all states reachable from any state in S by following only ϵ -transitions, including the states in S themselves.
- **Reversal Operation:** For a string $w = w_1w_2 \dots w_n$, its reversal is $w^R = w_n \dots w_2w_1$. For a language L , the reversed language is $L^R = \{w^R \mid w \in L\}$.
- **Complement Machine:** For a DFA $D = (Q, \Sigma, \delta, q_0, F)$, its complement machine is $D' = (Q, \Sigma, \delta, q_0, Q \setminus F)$, which recognizes the complement language $\Sigma^* \setminus L(D)$.

Questions!

1. We know that every NFA has an equivalent DFA. However, the number of states in this resulting DFA may vary. What is the range of the possible number of states? Can it be strictly less than k (the number of states in the NFA)? Can it exceed 2^k ? Give an example of an automaton attaining each of the true bounds.
2. We have a clear deterministic process to convert any NFA into a DFA. But does this work backwards? Is there a uniquely defined, optimal *reverse* algorithm that takes a minimal DFA and systematically deconstructs it into a minimal NFA? Why or why not? What makes finding the minimal NFA an inherently harder problem?
3. We know that for any two sets of states $A, B \subseteq Q$ in an NFA, the ϵ -closure distributes over union: $E(A \cup B) = E(A) \cup E(B)$. But does it also distribute over intersection? That is, does $E(A \cap B) = E(A) \cap E(B)$ always hold true? If so, provide a formal set-theoretic proof. If not, can you construct a rigid counterexample (explicitly defining Q , A , B , and the ϵ -transitions) where the equality fails?
4. We know how to find the complement machine for a DFA, but does the same simple state-flipping trick work for an NFA? Prove rigorously, via a minimal counterexample, that constructing $N' = (Q, \Sigma, \delta, q_0, Q \setminus F)$ from an NFA N does not necessarily yield the complement language $\Sigma^* \setminus L(N)$. What specific property of the transition function δ causes this algebraic closure to break down here?

5. Suppose we have a regular language L recognized by a DFA. Is its reversal, L^R , guaranteed to be regular as well? Prove that it is. (*Hint: Using an NFA might make this much easier*).
6. Start by stating the Pumping Lemma formally. Now, what would happen if we reversed the order of the first two quantifiers ($\exists p \geq 1$ and $\forall s \in L$)? Try to explain how this would completely change the meaning of the lemma. What class of regular languages would actually satisfy this altered version?
7. We know that many proofs naturally focus on *pumping up* ($i > 1$ in the expression xy^iz). But what about the other direction? Can you construct an example of a non-regular language where pumping up does not easily yield a contradiction, but *pumping down* ($i = 0$, resulting in the string xz) immediately breaks the structural rules of the language?
8. We know the Pumping Lemma is a necessary but not sufficient condition for a language to be regular. This means there are non-regular languages that can actually pass the test! Consider the language $L = \{a^ib^jc^k \mid i, j, k \geq 0 \text{ and if } i = 1 \text{ then } j = k\}$. First, can you prove that L is non-regular? Once you have that, demonstrate how it can somehow be successfully pumped regardless of how the adversary splits the string.
9. Consider the relationship between subsets and the Pumping Lemma. Prove or disprove the following statement:
"If $L_1 \subset L_2$ and L_2 satisfies the conditions of the Pumping Lemma, then L_1 must also satisfy the conditions of the Pumping Lemma because any string pumped in L_1 operates within the bounded structural rules of the parent language L_2 ."
10. Finally, let's look at the regularity of a few specific languages. Are the following languages regular? Provide proofs for your conclusions:
 - (a) $\{a^{n^2} \mid n \geq 0\}$
 - (b) $\{a^n b^m \mid n \leq m \leq 2n\}$
 - (c) $\{w \in \{0, 1\}^* \mid w = w^R\}$
 - (d) $\{w \in \{a, b\}^* \mid \text{count}(a) \equiv \text{count}(b) \pmod{3}\}$
 - (e) $\{w \in \{0, 1\}^* \mid w \text{ contains an equal number of the substrings } 01 \text{ and } 10\}$